

CAPE-Based Asset Allocation Strategy

Investment Research Project

Developed a dynamic asset allocation strategy using the Shiller CAPE ratio to evaluate long-term market valuation and optimize portfolio performance.

1. Introduction

The Cyclically Adjusted Price-to-Earnings ratio (CAPE), popularized by Robert Shiller, smooths reported earnings over a rolling ten-year window to produce a valuation measure that is less distorted by short-term earnings volatility than the standard trailing P/E ratio. Because valuation extremes have historically been associated with subsequent mean reversion in equity returns, CAPE is widely used as a long-horizon signal for assessing whether the broad market is cheap or expensive relative to history.

This project uses annual S&P Composite data from 1922 through 2024 — prices, dividends, earnings, CPI, and Treasury yields — to construct a real (inflation-adjusted) CAPE series and to test whether that series has predictive power for future equity returns. The analysis then translates this signal into a simple, rules-based dynamic asset allocation strategy that shifts the equity/Treasury-bill mix in response to the prior year's CAPE reading, and benchmarks the resulting performance against a static 60/40 portfolio.

2. Methodology

2.1 Data and CAPE Construction

Real CAPE in a given year is computed as the inflation-adjusted S&P Composite price index divided by the trailing ten-year average of inflation-adjusted earnings. Real and nominal annual price returns, along with forward 1-year and forward 5-year returns (both nominal and real), are derived from the price, dividend, and CPI series to evaluate the predictive relationship between valuation and subsequent performance.

2.2 Regression Analysis

Four single-variable OLS regressions were run, regressing forward 1-year and forward 5-year returns (nominal and real) on the prior year's real CAPE ratio. In every specification the CAPE coefficient is negative, consistent with the mean-reversion hypothesis that higher valuations are associated with lower subsequent returns.

For 1-year horizons, the relationship is weak: the real-return regression has an R^2 of roughly 3% and the CAPE coefficient is not statistically significant at the 5% level ($p \approx 0.085$), implying that short-term returns are dominated by noise rather than valuation. For 5-year horizons, the relationship strengthens considerably: R^2 rises to roughly 13% (real) and 25% (nominal), with CAPE coefficients significant at well beyond the 1% level. This is consistent with the view that valuation-driven mean reversion operates over a multi-year horizon rather than from one year to the next.

2.3 Dynamic Allocation Strategy

Using the prior year's real CAPE as the signal, the portfolio is rebalanced annually according to three valuation bands:

- CAPE below 15 (undervalued): 80% S&P 500 / 20% Treasury bills
- CAPE between 15 and 25 (fair value): 60% S&P 500 / 40% Treasury bills
- CAPE above 25 (overvalued): 30% S&P 500 / 70% Treasury bills

This dynamic strategy is compared against a static 60/40 benchmark held constant across the full sample period.

3. Results

Across the historical sample, the CAPE-based dynamic strategy produced an average annual return of approximately 4%, compared to approximately 3% for the static 60/40 benchmark. The dynamic strategy outperformed in most years when valuations were at extremes — increasing equity exposure after periods of depressed valuations (e.g., the 1930s and early 1980s) and reducing exposure ahead of richly valued, lower-return periods (e.g., the late 1990s and late 2010s) — while tracking the benchmark closely when CAPE sat in the middle of its historical range.

The performance gap is modest in absolute terms, but it is achieved without taking on additional realized volatility, since the strategy reduces equity exposure precisely when valuations — and historically, forward volatility — tend to be elevated.

4. Conclusion

The evidence supports the core premise of valuation-based asset allocation: CAPE has limited power to predict next year's return but meaningfully more power to predict returns over a five-year horizon, where the regression relationship is both economically and statistically significant. Translating this signal into a simple three-tier rebalancing rule modestly improved risk-adjusted performance relative to a static 60/40 portfolio over the 1922–2024 sample.

The strategy's limitations are worth noting. It relies on a single valuation metric, ignores transaction costs and taxes, uses fixed (rather than continuous) allocation bands, and is tested on overlapping historical data that may not fully capture structural shifts in market valuation over the past century. Extensions could include continuous (rather than discrete) position sizing as a function of CAPE, the addition of complementary signals such as momentum or interest-rate trends, and out-of-sample or rolling-window validation to test robustness.

Executive Summary

This project develops and backtests a dynamic, CAPE-based asset allocation strategy using 103 years of S&P Composite data (1922–2024), and compares it against a static 60/40 equity/Treasury-bill benchmark.

Key Findings

- CAPE exhibits stronger predictive power for 5-year returns ($R^2 \approx 13\text{--}25\%$, statistically significant) than for 1-year returns ($R^2 \approx 3\text{--}6\%$, not statistically significant), consistent with valuation-driven mean reversion operating over multi-year horizons.
- The dynamic, CAPE-based allocation strategy modestly outperformed the static 60/40 benchmark over the historical sample, generating an average annual return of approximately 4% versus approximately 3%.
- Lower CAPE readings (undervaluation) were generally followed by higher subsequent returns, while higher CAPE readings (overvaluation) were generally followed by lower subsequent returns, supporting the use of valuation as a long-horizon allocation signal rather than a short-term timing tool.
- The strategy's edge is concentrated at valuation extremes; in the middle CAPE range (15–25) performance closely tracked the static benchmark, suggesting the greatest practical value lies in adjusting exposure at cycle extremes rather than fine-tuning allocations continuously.